Question:

A silicious rock contains the mineral ZnS. To analyze for Zn, a sample of the rock is pulverized and treated with HCl to dissolve the ZnS (silicious matter is insoluable). Zinc is precipitated from solution by the addition of potassium ferrocyanide K_4 Fe (CN)_6. After filtering, the precipitate is dried and weighed. The reactions which occur are ZnS + 2HCl \rightarrow ZnCl_2 + H_2 S 2ZnCl_2 + K_4 Fe (CN)_6 \rightarrow Zn_2 Fe (CN)_6 + 4 KCl If a 2 gram sample of rock yields 0.969 gram of Zn_2Fe(CN)_6, what is the percentage of Zn in the sample? Atomic weight Zn = 65.4, molecular weight Zn_2 Fe (CN)_6 = 342.6.

Answer Choices:

A. 10 % Zn

B. 22 % Zn

C. 20 % Zn

D. 19 % Zn

E. 15 % Zn

F. 18 % Zn

G. 16 % Zn

H. 14 % Zn

I. 12 % Zn

J. 25 % Zn

Reasoning + Answer:

</tool_call> reasoning process here

We are given:

- A 2 gram sample of rock yields 0.969 g of ZnâĤĤFe(CN)âĤĨ.

- We need to find the percentage of Zn in the original rock sample.

Step 1: Use the molecular weights to find how much Zn is in the precipitate of ZnâĤĤFe(CN)âĤĨ.

Molecular weight of ZnâĤĤFe(CN)âĤĨ = 342.6 g/mol

Atomic weight of Zn = 65.4 g/mol

So, 2 moles of Zn in ZnâĤĤFe(CN)âĤĨ weigh:

2 ÃĹ 65.4 = 130.8 g of Zn

So, in 342.6 g of ZnâĤĤFe(CN)âĤĨ âĩĴ 130.8 g of Zn

Thus, the fraction of Zn in ZnâĤĤFe(CN)âĤĨ = 130.8 / 342.6

Now, the mass of Zn in 0.969 g of ZnâĤĤFe(CN)âĤĨ is:

Zn mass = (130.8 / 342.6) ÃĹ 0.969

Calculate that:

First, 130.8 / 342.6 âīĪ 0.3817

Then, 0.3817 ÃĹ 0.969 âīĪ 0.3705 g of Zn

This is the amount of Zn in the original 2 g rock sample.

Now, percentage of Zn in the rock = (mass of Zn / mass of sample) ÃĹ 100

= (0.3705 / 2) ÃĹ 100

= 18.525%

Closest value among the choices is:

F. 18 % Zn

<answer>18</answer>